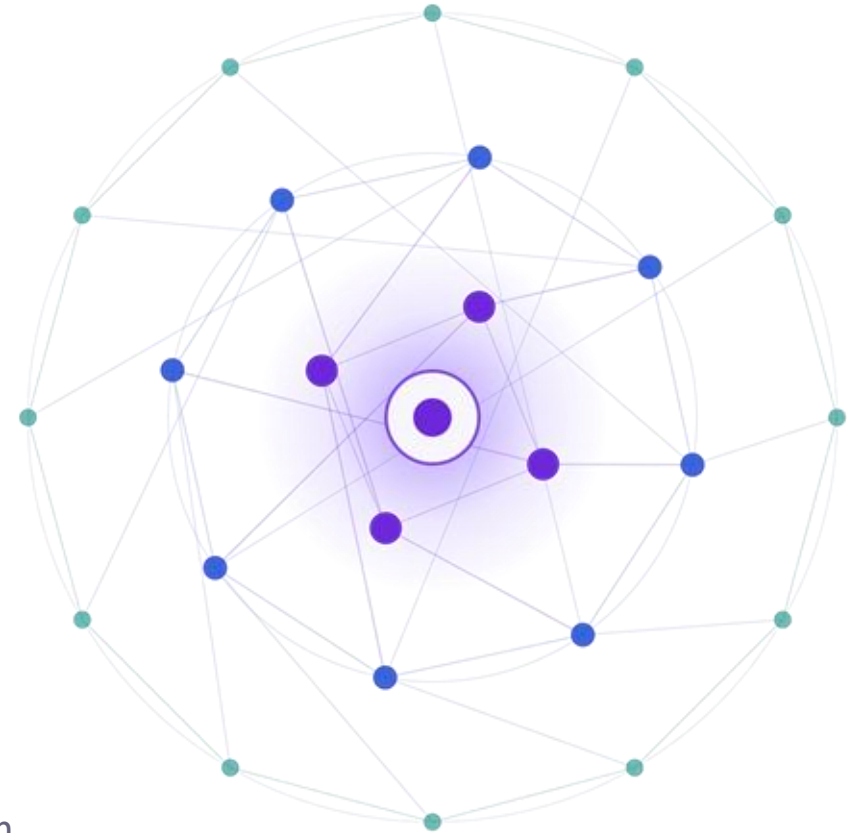


# Generative AI Hackathon

**3rd Edition** 2026 – 27

- Theme ID - 05
- Team Name - Caffeine
- College Name - SRMIST
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- Member Name & Email 4- Abhinav Prakash – abhinavprakash355@gmail.com
- Submission Github link - <https://github.com/sahaj24/samsung-prism-test>



# Theme

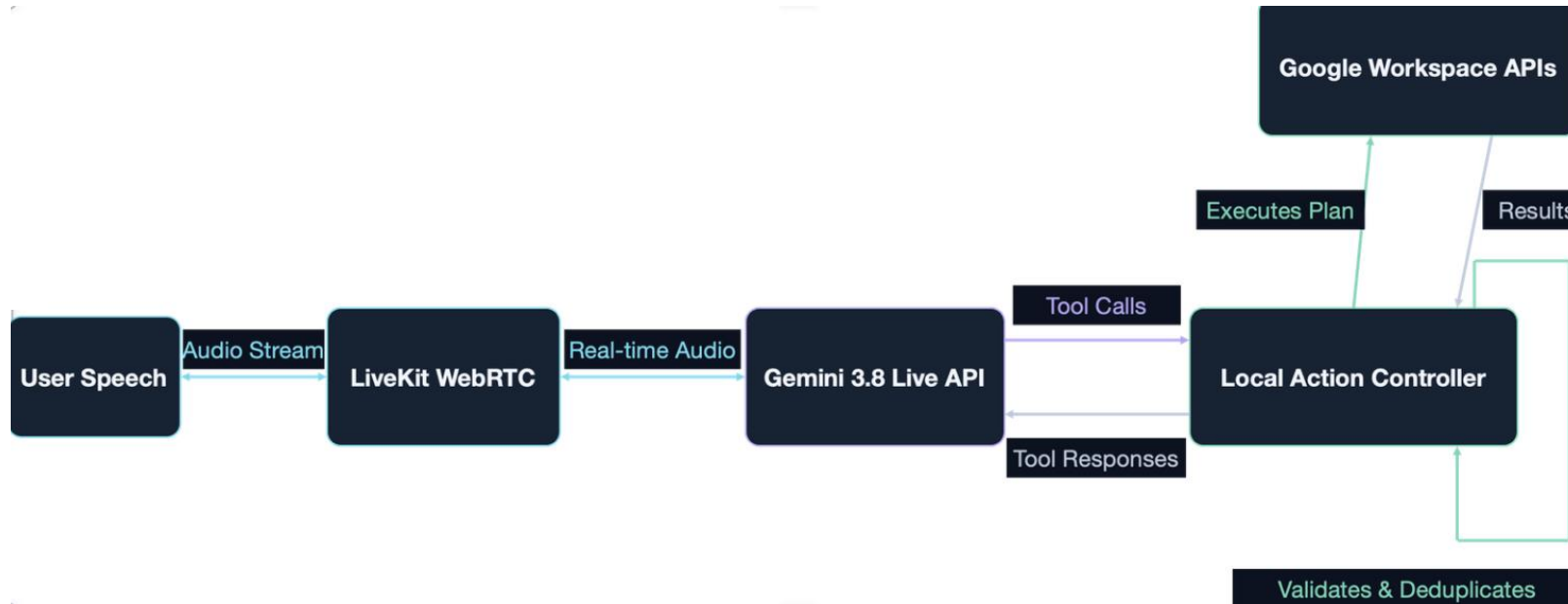
## Theme 05: Interruptible Real-Time Agents

- Reprise, Gemini Live and LiveKit support spoken requests and changes of mind.
- A controller checks tool arguments before dispatch and suppresses duplicate actions.
- Full-Duplex-Bench v3 tests tool-task completion on 100 human recordings.

# Existing Solutions & Gaps

- Voice models can propose tools, but speech interruption can leave actions running.
- Corrections can invalidate queued calls or change arguments before dispatch.
- Multi-step tasks need IDs from real tool results, rather than guessed values.
- Duplicate writes and uncertain outcomes need explicit execution tracking.

# Our Solutions & Architecture Diagram



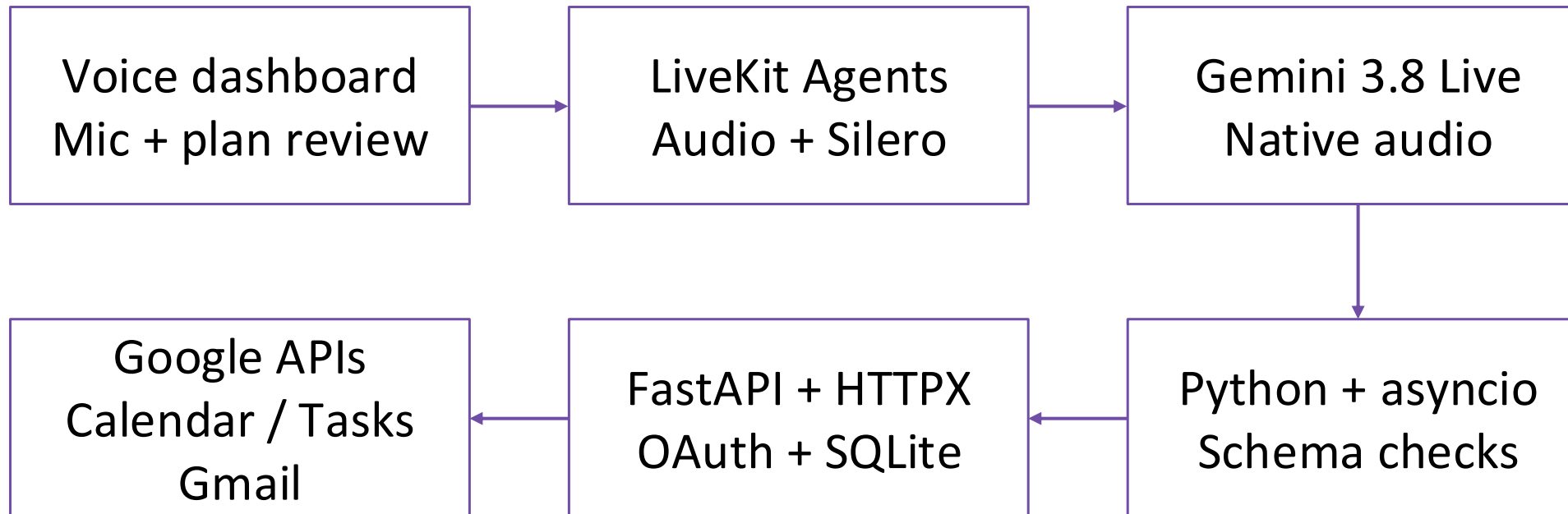
Revision and schema checks reject stale or invalid calls.  
Google writes require plan confirmation.

# Demo & Product Walkthrough

1. Start the voice worker and local dashboard, then enable the microphone.
2. Speak a request and correct it before dispatch. Inspect the tool trace.
3. Workspace demo: connect Google, request an event, task, draft and shopping list.
4. Review the corrected plan, confirm it and inspect returned Google IDs.

Recorded demo: 3:04, from released-v8 (74/100). Current benchmark: 77/100.

# Tools and tech stack used



Benchmark path: controller uses the public benchmark tool adapters.

Evaluation: Full-Duplex-Bench v3, FFmpeg, NumPy, SoundFile

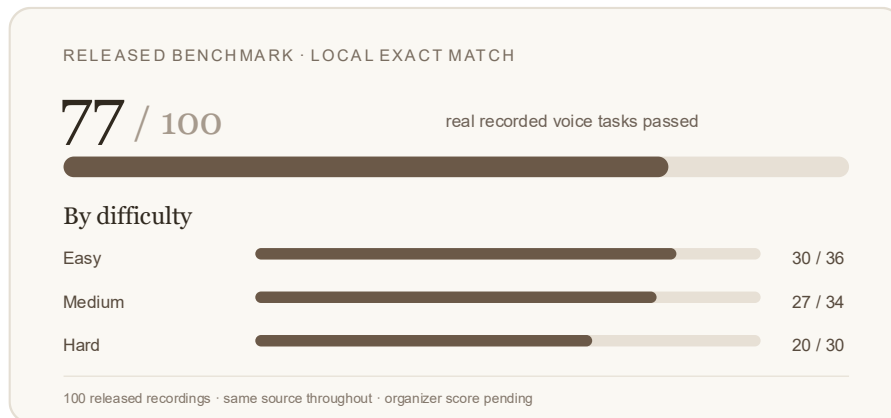
Development: uv, pytest, pytest-asyncio, Ruff

# Impact & Use case

- Voice assistance: handle corrected requests while tracking tool execution.
- Daily planning: turn a spoken request into an event, task, email draft and checklist.
- User control: preview Workspace actions and require confirmation before execution.
- Evidence: 77/100 local benchmark passes. Live Google writes still need validation.

# Innovation highlights, results and limitations

## Results:



- **Innovation:**
- Revision-aware dispatch, validated arguments and duplicate suppression.
- 100/100 recordings completed. Two ended in infrastructure failure.
- **Limits:**
- Uncertain responses: A lost API response can leave an action's outcome unknown. We avoid automatically repeating it.

# What's next

- Validate Google writes after OAuth setup and confirm returned resource IDs.
- Inspect failed benchmark cases and compare guarded versus baseline runs.
- Measure correction latency on the current frozen version.
- Test harder multi-step requests and recovery from partial failures.

# Brownie points slide( differentiation)

- Real human audio: All 100 released recordings remain in the score denominator.
- Reproducible evidence: Frozen source, pinned benchmark and per-case tool traces.
- Execution tracking: Dispatched writes remain recorded after interruption.
- Workspace control: Corrected previews need fresh confirmation before execution.

# Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo (Y/N) - Y
- README with reproducible setup instructions (Y/N) - Y
- Demo video, max 5 minutes (YouTube or Drive link) - Y
- Presentation file (PPT or PDF) (Y/N) - Y

# Thank you

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India